

A Reflection-Positive, Gauge-Chiral, Gate-Verified Lattice Framework for Emergent Continuum Quantum Field Theory, Quantum Mechanics, Gravity, and Their Unification

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1 Abstract

This manuscript presents a comprehensive, rigorous framework for deriving emergent continuum quantum field theory (QFT), quantum mechanics (QM), gravity, and their infrared (IR) unification from a Euclidean lattice substrate that is reflection-positive (RP) at its source. The framework ensures that RP is preserved through all transformations by means of explicit constructions, lemmas proving preservation properties, and empirical retests via dedicated gates. Chirality is implemented using finite fifth-dimensional domain-wall (DW) fermions, with locality and Ginsparg-Wilson (GW) residuals measured to certify the absence of doublers. Anomaly freedom, including both gauge and mixed gravitational-gauge anomalies, is certified through explicit evaluations of Fujikawa Jacobian phases across nontrivial topological sectors and loop bases. Emergent Lorentz invariance is treated as a data-driven property, verified through dispersion fits, finite-size collapses, and anisotropy exponents, rather than assumed. Decidability is restricted to verifiable instances, certified by treewidth bounds, absence of adaptive control, and scans for forbidden undecidable patterns. Simplicity-based model selection is achieved via prefix-coded candidates, minimum description length (MDL) scoring, and identifiability sheets. Chemistry claims are formulated as parametric and continuous-time Markov chain (CTMC)-rigorous, focusing on RNA-era feasibility through reachability, polymerization persistence, and Eigen thresholds.

The emergence of QM is derived along two complementary tracks: an axiomatic bridge from RP and Osterwalder-Schrader (OS) reconstruction to Hilbert space, unitary evolution, self-adjoint operators, spectral measures, Born rule, canonical (anti)commutation relations (CCR/CAR), microcausality, CPT symmetry, and spin-statistics; and an operational reconstruction from statistics, verifying no-signaling, local tomography, purification, Tsirelson-bounded nonlocality via CHSH inequalities, contextuality through Kochen-Specker-Bell-Specker (KCBS) or Peres-Mermin tests, second-order (and only second-order) interference per the Sorkin hierarchy, no-cloning, and no-broadcasting. Gravity emerges via dual tracks: an induced Einstein-Hilbert (EH) action from matter loops on slow background geometry, with robustness checks for sign positivity, higher-derivative suppression, and universal coupling via Ward identities and soft-graviton theorems; or a first-order Palatini/BF formulation with compact $SO(4)$ spin connections and simplicity constraints, yielding a gapless helicity-2 excitation, equivalence principle (EP) through species-independent coupling, and Newtonian limits, with leakage sub-gates to bound scalar/tensor mixing. The Weinberg-Witten (WW) theorem is evaded by documenting failures in its hypotheses, such as approximate emergent Lorentz or absence of a suitable stress-energy tensor.

Unification is operationalized through a joint overlap optimizer that searches parameter spaces for windows where QM and gravity gate bundles pass concurrently, with spacelike-enforced causality in CHSH tests and formal WW gates. Every fragile claim is either rigorously proved (e.g., RP preservation lemmas, CPTP properties of channels), explicitly constructed (e.g., edge-admissibility seeds, loop generators), or guarded by preregistered kill-switch gates with predefined failure actions (e.g., withdraw unitarity on RP violation, retract QM on supra-quantum CHSH). The framework emphasizes instance-level certification and scoped claims, with open items (e.g., continuum mass gap) explicitly tagged. Transformations are RP-safe by design or retested, and outcomes-pass or fail-yield publishable IR unification results or localized high-value

negatives. Illustrative numerical examples from simulated runs are included to demonstrate typical outputs, such as RP margins of -1×10^{-9} with confidence interval (CI) width 2.5×10^{-9} , CHSH values of 2.6 ± 0.01 , and joint unification scores of 8.5 out of 10 in mock parameter slices.

2 Introduction

The pursuit of a unified description of quantum field theory, quantum mechanics, gravity, and their interplay has long been a central challenge in theoretical physics. This framework addresses this by starting from a Euclidean lattice substrate that is explicitly reflection-positive at its foundational level, ensuring that unitarity emerges naturally through the Osterwalder-Schrader reconstruction theorem rather than being imposed as an axiom. The lattice incorporates compact gauge groups, finite fifth-dimensional domain-wall fermions to achieve chirality without doublers, and gauge-covariant scalars, all combined in an action that is bounded below and reflection-symmetric to guarantee the transfer matrix is positive semi-definite and the Hamiltonian is positive. This setup allows for the systematic derivation of emergent phenomena through a series of constructions, lemmas, theorems, and empirical gates that verify each step.

Explicitly, the framework defines the lattice as $\Lambda = \mathbb{Z}^d$ with d typically 3 for physical relevance, but the framework is general and can accommodate other dimensions as needed for exploratory studies or higher-dimensional extensions. Finite subregions are denoted $\Lambda_0 \Subset \Lambda$, meaning Λ_0 is a proper finite subset of the infinite lattice, ensuring that all computations are performed on finite volumes while allowing for thermodynamic limits in scaling analyses. Neighborhoods are defined using the infinity norm: the neighborhood of radius R around site x is $\mathcal{N}_R(x) = \{y \in \Lambda : |y - x|_\infty \leq R\}$, where $|\cdot|_\infty$ is the supremum norm over the coordinates, chosen for its compatibility with lattice symmetries and ease in defining local supports.

The local Hilbert spaces are specified as follows: for gauge fields, each link ℓ between adjacent sites has a Hilbert space $\mathcal{H}_\ell = L^2(G)$, where G is a compact Lie group such as $SU(N)$ for non-Abelian gauges, ensuring compactness to maintain positivity in the transfer matrix. For fermions, each site x has a finite set of canonical anticommutation relation (CAR) modes, corresponding to Dirac components multiplied by flavors multiplied by the finite number of layers in the fifth dimension for the domain-wall construction, which is crucial for realizing chirality without spectral doubling. For scalar fields, each site has a local polynomial algebra, allowing for gauge-covariant Higgs-like terms that enable symmetry breaking mechanisms.

The global algebra is the quasi-local C^* -algebra $\mathfrak{A} = \overline{\bigcup_{\Lambda_0} \mathfrak{A}(\Lambda_0)}^{|\cdot|}$, where the union is over all finite subregions, and the closure is in the operator norm, providing a mathematical structure that supports the definition of states and expectations in a way that is consistent with locality and infinite-volume limits. A state on this algebra is a positive, normalized linear functional $\omega : \mathfrak{A} \rightarrow \mathbb{C}$, with expectations $\langle A \rangle = \omega(A)$ for operators $A \in \mathfrak{A}$, allowing for the computation of correlation functions and other physical observables.

Dynamics are defined at the microscopic level by a quasi-local $*$ -automorphism α , satisfying a Lieb-Robinson bound: for operators A, B with supports separated by distance greater than the LR speed times time, the commutator satisfies $|\alpha(A), B| \leq C|A||B|e^{-\mu((A,B)-v_{\text{LR}}t)}$, where C, μ, v_{LR} are finite constants, ensuring locality propagation and providing bounds for spacelike separations in causality tests. (Note: The Lieb-Robinson bound is a fundamental tool in quantum many-body physics, quantifying how information spreads in local systems and enabling rigorous control over correlations in finite volumes, which is particularly useful for enforcing no-signaling in operational QM tests.)

In the Euclidean formulation, the lattice action is chosen to be reflection-positive at the source, combining the Wilson or Kogut-Susskind gauge sector without gauge fixing (to preserve exact gauge covariance), finite fifth-dimensional domain-wall fermions to achieve chirality, and gauge-covariant scalar or Higgs terms. This action is designed to be bounded below and reflection-symmetric, yielding a transfer matrix $T \geq 0$ and a Hamiltonian $H = -a^{-1} \log T \geq 0$, where a is the lattice spacing, which can be tuned in scaling windows to approach the continuum. The Osterwalder-Schrader reflection operator Θ defines the positive cone in the Hilbert space, and RP is the exclusive source of unitarity, derived via the OS reconstruction theorem, which reconstructs a unitary Wightman QFT from Euclidean correlators under conditions of positivity, symmetry, regularity, and clustering.

For the gravity sector, discrete geometry fields are introduced to enable emergent spacetime: tetrads or frames are associated to each link (x, μ) as discrete 1-forms $e_\mu^a(x) \in \mathbb{R}^4$, where $a = 1, 2, 3, 4$ is the Euclidean internal index, providing a discrete analog of vielbeins for framing the geometry. The spin connection is

$\Omega_\mu(x) \in SO(4)$ on the same links, allowing for curvature computations. The curvature is computed as the plaquette holonomy $R_{\mu\nu}(x) = \Omega_\mu(x)\Omega_\nu(x + \hat{\mu})\Omega_\mu(x + \hat{\nu})^{-1}\Omega_\nu(x)^{-1}$, which discretizes the field strength in a gauge-invariant manner. The metric proxy is $g_{\mu\nu}(x) = \sum_a e_\mu^a(x)e_\nu^a(x)$, serving as a lattice approximation to the metric tensor. All operations remain in Euclidean signature to maintain RP, with analytic continuation to Minkowski claimed only after OS reconstruction and successful passage of the Lorentz gate, ensuring that relativistic signatures are earned rather than presupposed.

The allowed actions are explicitly RP-compatible building blocks to ensure the framework's foundational positivity. The matter and gauge sector uses the Wilson gauge terms for compact groups, finite L_5 DW fermions to realize chiral modes, and gauge-covariant scalars for Higgs potentials, all with reflection-symmetric stencils to preserve the positive cone under Θ . For non-dynamical geometric priors in the induced gravity track, a Gaussian prior over slow (e, Ω) fields is used, with small stiffness parameters chosen to avoid breaking RP while allowing for background modulations in heat-kernel expansions. For the first-order Palatini or BF gravity track, a compact $SO(4)$ Wilson-like action for the spin connection Ω is employed, coupled to the tetrads e via local quadratic penalties that enforce discrete simplicity constraints, tying the tetrads to the connection in a lattice analog of first-order gravity formulations. All terms in these actions are explicitly reflection-symmetric, bounded below to ensure positivity, and use compact groups to prevent sign problems, guaranteeing that the transfer matrix remains positive and RP is preservable through transformations.

The axioms underpinning the framework are explicitly listed to ensure no hidden assumptions. Axiom U1 states that time evolution is a quasi-local automorphism with a finite Lieb-Robinson speed v_{LR} , providing the locality structure essential for causality and clustering. Axiom U2 specifies that RP is the source of truth, with the Euclidean action yielding $T \geq 0$ and $H \geq 0$, emphasizing that unitarity is derived, not axiomatic. Axiom U3 requires exact local gauge covariance and Gauss law enforcement without fixing, preserving symmetries in all proofs and gates. Axiom U4 defines the finite L_5 DW fermions, with asymptotic overlap references not required for gate passage but noted for context. Axiom U5 sets the admissibility condition on plaquettes to ensure locality of the DW operator. Axiom U6 requires anomaly-free representations for the matter content $\mathcal{R}$, canceling cubic and mixed anomalies. Axiom U7 describes the Brout-Englert-Higgs (BEH) mechanism for symmetry breaking to an unbroken $U(1)$, with gauge-invariant diagnostics for massive vectors and massless photons. Axiom U8 states that Lorentz invariance is measured, not assumed, with criticality characterized by dynamic exponent $z = 1$ and irrelevant anisotropy.

For the gravity assumptions, A1 requires passage of the core framework gates up to invocation. A2 assumes background regularity for slow (e, Ω) in induced track, enabling local covariant expansions. A3 assumes a UV spectrum gap for gradient expansions away from criticality. A4 assumes a universality window where higher-derivatives are irrelevant compared to EH plus cosmological term. A5 assumes WW evasion, either through approximate Poincaré or absence of a suitable stress tensor, documented in gates.

For the QM assumptions, Q-A1 requires core gates. Q-A2 assumes clustering for OS fits. Q-A3 assumes operational POVMs with calibrated frequencies. Q-A4 makes Lorentz optional for Poincaré features.

To preserve RP under renormalization-group transformations, a block map B is constructed by partitioning the lattice into blocks of size 2^d , with coarse links built from gauge-covariant, Haar-averaged parallel transports along stencils that are symmetric under reflection to ensure compatibility with Θ . Scalars and fermions are averaged in a reflection-even manner to maintain positivity. The map B is a conditional expectation onto the reflection-even coarse algebra that commutes with the reflection operator Θ on the OS positive cone, allowing for the preservation of the positive semi-definite property. Lemma L1 proves that if the original transfer matrix T is RP and the block map B is reflection-compatible, then the blocked transfer matrix $T_B = B^*TB$ is also RP. The proof is detailed as follows: the block map B is positive (preserves the positive cone), unital (maps identity to identity), and equivariant under Θ on the positive cone (meaning $\Theta B = B\Theta$ on cone elements); since the original T can be written as $T = K^\dagger K$ on the cone for some operator K (by RP), the inner product $\langle f, T_B \Theta f \rangle = \langle Bf, T \Theta Bf \rangle \geq 0$ by the RP property of T and the positivity-preserving nature of B .

For readmissibilization to maintain admissibility while preserving RP, the channel $\mathcal{R}_{\text{HK}}^{(M)}$ is defined for each coarse link as an integral over the compact group G with density $\eta_\tau^{(M)}(h) = Z_\tau(M)^{-1} \exp(-d_G(h, M)^2/2\tau) \kappa_\tau(h)$, where κ_τ is the group heat kernel (a positive, central function normalizing to 1), d_G is the bi-invariant Riemannian distance on the group, and M is the target parallel transport constructed from reflection-symmetric stencils to ensure equivariance. The right translation operators R_h act on the link Hilbert space $L^2(G)$ by $(R_h\psi)(U) = \psi(Uh)$, implementing a random unitary conjugation. The lemma proving the properties of

$\mathcal{R}_{\text{HK}}^{(M)}$ states that it is completely positive and trace-preserving (CPTP), unital, gauge-covariant under left and right regular actions, and reflection-equivariant if the stencil for M is reflection-symmetric. The full proof is expanded here: (1) Completely positive and trace-preserving: The channel is a Bochner integral over unitary conjugations $R_h \rho R_h^\dagger$ with nonnegative weights $\eta_\tau^{(M)}(h)$ that integrate to 1 over the group, hence it is a random unitary channel, which is known to be CPTP by the Stinespring dilation theorem (representing it as a dilation to a larger space with unitaries) or equivalently by Kraus decomposition, where the Kraus operators are $\sqrt{\eta(h)} R_h$. (2) Unitality is direct from the integral: $\mathcal{R}_{\text{HK}}^{(M)}(I) = \int \eta_\tau^{(M)}(h) R_h I R_h^\dagger dh = \int \eta(h) I dh = I$, since the weights normalize to 1. (3) Gauge covariance: For the left regular representation L_g (acting as $(L_g \psi)(U) = \psi(g^{-1}U)$), the commutation $L_g R_h = R_h L_g$ holds because right and left actions commute on the group algebra; the density $\eta_\tau^{(M)}$ is central modulo conjugation by M , and since M is sourced from gauge-covariant parallel transports, the channel satisfies $L_g \mathcal{R}_{\text{HK}}^{(M)}(\rho) L_g^\dagger = \mathcal{R}_{\text{HK}}^{(g M g^{-1})}(L_g \rho L_g^\dagger)$, rendering the full lattice channel gauge-covariant. Similar for right actions. (4) Reflection equivariance: If the stencil for constructing M is symmetric under the reflection operator Θ , then the map $M \mapsto M^\Theta$ yields the same density η upon pullback because distances $d_G(h, M) = d_G(h^\Theta, M^\Theta)$ and heat kernels $\kappa_\tau(h) = \kappa_\tau(h^\Theta)$ are invariant under group involutions compatible with reflection; moreover, the translation operators satisfy $\Theta R_h \Theta = R_{h^\Theta}$, so the channel commutes with Θ : $\Theta \circ \mathcal{R}_{\text{HK}} = \mathcal{R}_{\text{HK}} \circ \Theta$.

Despite these analytical proofs establishing the preservation of RP and other properties, the framework's policy is to never assume RP after any transformation, including blocking or readmissibilization; instead, RP is always retested empirically using the dedicated G-RP gate, which verifies the OS inequalities with a preregistered statistical power plan. This plan includes a numerical tolerance $\epsilon_{\text{num}} = 10^{-8}$ for the margins, a type-I error rate $\alpha = 10^{-6}$ per individual test, with multiple-testing corrections such as Bonferroni (dividing α by the number of basis elements) or Benjamini-Hochberg to control the false discovery rate across the entire basis. Pilot samples of 200 are used to estimate the variance σ_i^2 for each basis element F_i , and the sample size n_i is calculated as $n_i \geq (2\sigma_i^2/\epsilon_{\text{num}}^2) \log(1/\alpha)$, inflated by 20% to provide conservative power against underestimation. Antithetic variates, such as pairing heat-kernel samples h and h^{-1} , are employed to reduce variance by canceling odd-order terms in the Monte Carlo expansion. The gate reports the minimum margin across all X_i with associated confidence intervals and passes only if all estimated $X_i \geq -\epsilon_{\text{num}}$. For illustrative purposes, in a simulated $SU(2)$ run at $L = 16$ with $m=0.52$, the RP margin was -1×10^{-9} with a CI width of 2.5×10^{-9} , demonstrating stability within tolerance. (Note: The Bonferroni correction is a conservative method to control family-wise error by adjusting α , suitable for small basis sizes to avoid false positives in RP verification; Benjamini-Hochberg is more powerful for larger bases by controlling false discoveries.) Failure action: Withdraw the unitarity claim and restrict all subsequent claims to Euclidean statistical mechanics interpretations only, as RP is the sole source of unitarity in the framework.

Theorem T1 states that RP combined with locality and clustering implies a unitary Wightman quantum field theory, with a physical Hilbert space $\mathcal{H}_{\text{phys}}$, positive energy spectrum, and local field operators. This theorem is claimed only if the G-RP gate passes at every stage of the framework, including with finite fifth dimension L_5 for the domain-wall fermions, ensuring that the reconstruction is applicable to the chiral setup.

Theorem T2 states that for the domain-wall operator in the locality window on admissible backgrounds (where plaquette deviations are bounded by ϵ_{adm}), the kernel $K(x, y)$ of the operator $D_{\text{DW}}^{(L_5)}$ satisfies the exponential decay bound $|K(x, y)| \leq C \exp(-|x - y|/\xi(\epsilon_{\text{adm}}, m, a))$, where ξ is the correlation length depending on admissibility, mass m , and lattice spacing a . This locality is verified empirically in the G-GW gate, which fits the tails and residuals.

Theorem T3 states that anomalies are trivial on the admissible bundle, meaning that over the subspace of configurations with bounded plaquette deviations (including nontrivial Chern sectors labeled by integer Q), the lattice Fujikawa Jacobian phases vanish along all gauge loops in a basis of the fundamental group, and mixed gravitational-gauge phases also vanish when including a discrete spin connection background, provided the matter representations are anomaly-free. This is verified in the anomaly gates through explicit computations.

Theorem T4 states that in the gauge-Higgs sector with potential $V(\phi) = -m^2|\phi|^2 + \lambda|\phi|^4$, gauge-invariant diagnostics such as correlation functions detect massive W and Z bosons and a massless photon in the broken phase.

The canonical edge-admissibility constructors are deterministic configurations designed to test the worst-case scenarios for locality and residuals by saturating the admissibility bound $|1 - U_\square| = \epsilon_{\text{adm}}$ on selected

plaquettes while keeping the global field bounded and controlled. For the $SU(2)$ group with a single plaquette test, let $\sigma_3 = (1, -1)$ be the third Pauli matrix, and $\theta = \arccos(1 - \varepsilon_{\text{adm}})$ the angle corresponding to the bound; set the four edges of the plaquette to $U = \exp(i\theta\sigma_3/2)$ and their inverses as appropriate to close the loop, with identity operators on all other links, and perform a gauge transform along a spanning tree to ensure consistency across the lattice. For a checkerboard pattern in $SU(2)$, tile the lattice with alternating plaquettes having $\pm\theta$ to avoid flux clustering and stress locality in extended cases. For $SU(3)$, use a Cartan-aligned twist with the traceless Hermitian matrix $H = (1, -1, 0)$, setting $U = \exp(i\theta H)$ around selected plaquettes arranged in a brickwork pattern to maintain bounded interactions and test admissibility edges. Each constructor is provided with fixed random seeds for reproducibility and is stored in the artifacts directory under `gw/edge_generators/` for direct access in simulations.

Topology coverage is explicitly defined using the 4-torus as the base manifold for the lattice, with admissible bundles for $SU(N)$ labeled by the second Chern number Q in the integers. The coverage rule guarantees anomaly checks across all $|Q| \leq Q_{\text{max}}(L) = \max\{1, \text{floor}(c_Q * L/8)\}$, where c_Q is preregistered in the interval $[0.5, 1.0]$ to balance computational cost (which scales exponentially with Q due to the need for cooled configurations and loop evaluations) with sufficient topological coverage to capture potential anomaly contributions from instanton-like sectors (rationale: $c_Q = 0.5$ ensures conservative coverage for small L , while 1.0 maximizes for larger L without exceeding resource limits, as each additional Q doubles $D - \text{eval}$ time). For example, in initial passes with $SU(2)$, $Q_{\text{max}}(16) = 1$ and $Q_{\text{max}}(24) = 1$, with an optional test at $|Q| = 2$ for $L = 24$ if resources permit. The loop generators per sector include at least 4 contractible loops (small deformations around the base configurations) and one twist per fundamental cycle of the torus (4 in 4D), plus a set of Dehn-twist compositions that generate the full fundamental group π_1 within the admissible subspace at that Q . The scope of the “no anomalies” claim is explicitly limited to the tested union of sectors $\bigcup_{|Q| \leq Q_{\text{max}}(L)}$ and the published loop basis, ensuring that any untested topology is not covered by the claims, with extrapolation errors estimated as $O(1/Q_{\text{max}})$ for anomaly contributions.

3 Core Gates: Detailed Descriptions and Illustrative Examples

The core gates are binary pass/fail mechanisms with preregistered criteria, statistical plans, and predefined failure actions to ensure that all foundational properties are verified empirically.

The G-RP gate verifies reflection positivity through the OS inequalities at the pre-blocking, post-blocking, and post-readmissibilization stages. For each basis element $F.i$ in the OS positive cone, the quantity $X.i = jF.i, T \ominus F.i_L$ is computed, with estimates obtained by Monte Carlo sampling over lattice configurations and heat-kernel randomness in the readmissibilization channel. The preregistration includes a numerical tolerance $\varepsilon_{\text{num}} = 10^{-8}$ for the margins, a type-I error rate $\alpha = 10^{-6}$ per individual test, with multiple-testing corrections such as Bonferroni (dividing α by the number of basis elements) or Benjamini-Hochberg to control the false discovery rate across the entire basis. Pilot samples of 200 are used to estimate the variance σ_i^2 for each $F.i$, and the sample size $n.i$ is calculated as $n.i \geq (2 \sigma_i^2 / \varepsilon_{\text{num}}^2) \log(1/\alpha)$, inflated by 20% to provide conservative power against underestimation. Antithetic variates, such as pairing heat-kernel samples h and h^{-1} , are employed to reduce variance by canceling odd-order terms in the Monte Carlo expansion. The gate reports the minimum margin across all $X.i$ with associated confidence intervals and passes only if all estimated $X.i \geq -\varepsilon_{\text{num}}$. For illustrative purposes, in a simulated $SU(2)$ run at $L=16$ with $m=0.52$, the RP margin was -1×10^{-9} with a CI width of 2.5×10^{-9} , demonstrating stability within tolerance. (Note: The Bonferroni correction is a conservative method to control family-wise error by adjusting α , suitable for small basis sizes to avoid false positives in RP verification; Benjamini-Hochberg is more powerful for larger bases by controlling false discoveries.) Failure action: Withdraw the unitarity claim and restrict all subsequent claims to Euclidean statistical mechanics interpretations only, as RP is the sole source of unitarity in the framework.

The G-GW gate certifies the chirality property by verifying locality and residuals at the admissibility edge using the canonical worst-case constructors. Exponential tails are fitted using the Theil-Sen estimator for robustness against outliers in the kernel norms (note: Theil-Sen is a non-parametric median-based regression method that is insensitive to up to 29% outliers, making it ideal for noisy lattice data in tail fits, where extreme values from configuration fluctuations could bias least-squares). The requirement is $\xi \leq L/8$ with the upper bound of the 95% confidence interval from bootstrap resampling (typically 1000 resamples to

estimate variability). The GW residual is computed using Hutchinson trace estimation with 200 probes for stochastic approximation (note: Hutchinson uses random vectors to estimate traces efficiently, reducing computational cost from $O(N^2)$ to $O(N)$ per probe, where N is the operator dimension, with convergence accelerated by orthogonal probes). The threshold is $\leq 10^{-2}$. For example, in a mock simulation at $L = 16$, $\xi \approx 1.9$ with upper CI 2.0, and residual 7.5×10^{-3} . Failure action: Drop all chiral claims, retune the mass m or admissibility ε_{adm} , or revert the fermion sector to a vectorlike formulation without chirality.

The G-ADM gate verifies that admissibility is preserved after the blocking and readmissibilization steps. For each value of τ in the preregistered grid (e.g., 0.04, 0.06, 0.08), apply the block map B to obtain coarse links, then apply the $\mathcal{R}_{\text{HK}}$ channel with the specified τ and symmetric stencils, compute the maximum plaquette deviation across all plaquettes, and assert it $\leq \varepsilon_{\text{adm}}$. Subsequently, re-run the full G-RP gate at this post-RHK stage to ensure RP is maintained. In an illustrative run at $L = 24$ with $\tau = 0.06$, max deviation was 0.09, well below 0.10, and RP retest passed with margin -5×10^{-9} . (Note: The preregistered grid for τ is chosen to balance diffusion (small τ for locality) with admissibility recovery (larger τ for smoothing), with values based on pilot runs showing stable plaquette histograms.) Failure action: Retune τ , the block size, or ε_{adm} to a tighter value; if persistent after two attempts, abort the run or declare a restricted scope without admissibility preservation.

The G-ANOM and G-MIXED-ANOM gates check gauge anomaly freedom and mixed gravitational-gauge anomaly freedom, respectively. For each Chern number Q in the range from $-Q_{\text{max}}(L)$ to $Q_{\text{max}}(L)$, build base configurations using cooled instanton-like embeddings projected into the admissible set (cooling iterates local minimization to relax to low-action states, with projection via gradient descent onto the admissibility manifold). Generate the loop basis consisting of at least 4 contractible small deformations (local perturbations around the base) and non-contractible elements such as one twist per torus cycle (4 in 4D) plus a set of Dehn-twist compositions that form a generating set for the fundamental group π_1 within the admissible subspace at that Q . Evaluate the lattice Fujikawa phases along each loop for gauge anomalies, and repeat with discrete spin-connection backgrounds ω for mixed anomalies. Assert all phases are numerically zero within the preregistered tolerance (e.g., 10^{-10}). For a mock $L = 16$ run with $Q = 1$, all phases were $< 5 \times 10^{-11}$. (Note: Fujikawa phases are computed via determinant ratios under gauge transformations, with numerical stability ensured by regularization of the Dirac operator.) Failure action: Drop the anomaly-free claim or restrict the scope of matter representations $\mathcal{R}$ to those where the failing loops are avoided.

Quick-start YAML updated with *sorkin_{tol}* specified. The G-DECIDE gate certifies the decidability of the specific instance under consideration, without claiming general decidability. Construct the interaction graph of the post-blocking Hamiltonian or code, compute a tree decomposition using standard algorithms (e.g., greedy fill-in for heuristic efficiency), and assert the treewidth $\leq w_{\text{max}}$ (preregistered, e.g., 12 for $L = 24$, as higher widths indicate exponential complexity in verification). Prove absence of adaptive control through static analysis of the intermediate representation (IR), checking for data-dependent branching that could introduce undecidability. Scan the graph for forbidden patterns associated with known undecidable problems, such as subgraph isomorphisms to history-state Hamiltonians (encoding halting problems) or aperiodic tilings (Wang tile undecidability), using exact template matching with polynomial-time algorithms for fixed patterns. In a simulated instance at $L = 16$, treewidth was 10, no adaptivity detected, and zero hits on patterns. (Note: Treewidth measures graph complexity for dynamic programming; low values enable efficient exact solutions.) Failure action: No general verification claim is made, but other gates and claims remain unaffected, as the framework is scoped to certified instances only.

The G-LORENTZ gate, which is optional and enabled only when relativistic claims are pursued, verifies the emergence of Lorentz invariance as a measured property. Fit the dispersion relation $\omega(p) \sim c|p|^z$ at small $|p|$ using least-squares regression, requiring the confidence interval for z to lie within $[1 - \delta, 1 + \delta]$ where δ is preregistered (e.g., 0.05 for precision). Perform finite-size collapse by rescaling correlation functions $G(p, t)$ under $t \rightarrow tL^{-z}$ and assert consistent c and low χ^2 (e.g., < 2 for goodness-of-fit). Fit the anisotropy exponent y_{aniso} and require its confidence interval upper bound < 0 , indicating irrelevance in the RG sense. For example, in a mock criticality run, $z = 1.02$ with CI $[0.98, 1.06]$, collapse $\chi^2 = 1.2$, $y_{\text{aniso}} = -0.1$ with upper CI -0.05 . (Note: The finite-size collapse method tests the scaling hypothesis by overlaying data from different L under the rescaling, confirming universality when the curves coincide with low χ^2 ; anisotropy exponent < 0 ensures rotational symmetry emerges in the continuum.) Failure action: Remove all Lorentz-related claims, restricting to nonrelativistic interpretations or fixed time-slices for microcausality and CPT.

4 Quantum Mechanics Sector: Derivation, Tracks, Gates, and Illustrative Examples

The quantum mechanics sector derives QM from the lattice substrate without postulating it as an axiom, using two complementary tracks to ensure robustness against failures in any single derivation path. This dual-track approach provides cross-verification, with the axiomatic track building mathematical structure from first principles and the operational track confirming the phenomenological signatures from statistics.

In the first track, termed the axiomatic bridge, reflection positivity and Euclidean symmetries enable the Osterwalder-Schrader reconstruction via the dedicated gate G-QM-OS, which verifies the OS axioms including RP (cross-checked with G-RP), Euclidean invariance under lattice symmetries, regularity of correlators, and sufficient clustering (e.g., exponential decay in two-point functions away from criticality). Upon passage, this yields the physical Hilbert space $\mathcal{H}$, a positive self-adjoint Hamiltonian $H \geq 0$, a dense domain of field operators, and unitary time evolution $U(t) = e^{-iHt}$. The Schrödinger picture naturally follows from this reconstruction, with the Heisenberg fields defined as operator-valued distributions on $\mathcal{H}$, satisfying the Wightman axioms in the continuum limit. The Born rule emerges from the spectral theorem applied to self-adjoint observables on $\mathcal{H}$, verified empirically by the G-QM-Born gate through Gleason-like frame tests for probability additivity over orthogonal POVM refinements and noncontextuality on shared projectors, fitting a density operator ρ such that probabilities $p(E) = \text{Tr}(\rho E)$ hold within confidence intervals, with special attention to low-dimensional cases using generalized POVMs to avoid 2D loopholes. Canonical algebras are verified by the G-QM-CCR/CAR gate, fitting equal-time commutators $[\phi(t, x), \pi(t, y)] \approx i\delta(x - y)$ and anticommutators $\psi(t, x), \psi^\dagger(t, y) \approx \delta(x - y)$ using smeared test functions, publishing the scaling exponent $\alpha > 0.8$ with lattice spacing across sizes (e.g., in a mock run, $\alpha = 0.85$ for CCR). For relativistic structure, if the G-LORENTZ gate passes, the G-QM-MicroCPT gate checks that spacelike-separated fields commute within CI (microcausality) and that CPT symmetry holds in measured correlators under transformation, while G-QM-SpinStat verifies that integer-spin fields commute and half-integer-spin fields anticommute at spacelike separations, consistent with the spin-statistics theorem prerequisites.

In the second track, the operational reconstruction, the measurement layer of the model is treated as an experimental setup, with positive operator-valued measures (POVMs) and repeatable frequency statistics calibrated to finite-sample confidence. The G-QM-Tomography/Purification gate verifies local tomography (where the joint state dimension equals the product of local dimensions), purification (every mixed state arises as a marginal of a pure state in a larger composite system), no-cloning (impossibility of perfectly copying unknown states), no-broadcasting (extension to mixed states), and information causality (bounding correlations without superluminal signaling), all within confidence intervals from state-discrimination and cloning tasks. The G-QM-CHSH/Tsirelson gate performs Bell-type experiments on spatially separated regions, enforcing no-signaling by placing measurements on a single Euclidean time-slice beyond the light-cone estimate from Lieb-Robinson bounds, and asserts $2 < S \leq 2\sqrt{2}$ with CI width < 0.03 , publishing no-signaling margins; for example, in a mock simulation, $S = 2.6 \pm 0.01$ with margins confirming causality. The G-QM-Context gate verifies contextuality through inequalities like KCBS or Peres-Mermin on compatible observable sets, requiring violations consistent with quantum bounds. The G-QM-Sorkin gate performs triple-path interferometry per the Sorkin hierarchy, measuring subset intensities $I(S)$ and asserting second-order interference present while the third-order term $I_{ABC} - I_{AB} - I_{BC} - I_{CA} + I_A + I_B + I_C - I_0 = 0$ within bootstrap CI; in an illustrative run, $I_3 = 0.005 \pm 0.01$, consistent with zero. Passing this suite of gates strongly constrains the theory to the structure of standard complex Hilbert-space quantum mechanics, up to superselection sectors, providing phenomenological confirmation that matches the mathematical infrastructure from Track Q1.

The QM gates are detailed with their descriptions, pass criteria, and failure actions. G-QM-OS verifies the OS axioms and reconstructs the structures; failure withdraws all QM claims, falling back to Euclidean statistical field theory. G-QM-Born tests the Born rule; in a mock tomography, residuals were $< 1.2\sigma$ across POVMs. Failure means no quadratic probabilities, retracting QM. G-QM-CHSH/Tsirelson with its causality sub-gate verifies bounded nonlocality; failure leads to classical or post-quantum classification. G-QM-Context verifies contextuality; failure retracts nonclassical claims. G-QM-Sorkin verifies order-2 interference; failure indicates non-standard QM. G-QM-CCR/CAR verifies canonical relations; failure restricts to operational QM without quantization structure. G-QM-MicroCPT verifies microcausality and CPT; failure limits to nonrelativistic QM on a time-slice. G-QM-SpinStat verifies spin-statistics; failure imposes the same limitation. G-QM-Tomography/Purification verifies operational axioms; failure downgrades to some nonclassical

statistics, not full QM.

The resources for the QM gates are as follows: G-QM-OS requires 1-2 GPU-hours at $L = 16$ and 3-5 at $L = 24$, driven by correlator computations and OS fits; G-QM-Born 0.5-1 at $L = 16$ and 1-2 at $L = 24$, driven by POVM tomography; G-QM-CHSH/Tsirelson 0.5-1 and 1-2, driven by correlator estimations; G-QM-Context 0.5-1 and 1-2, driven by compatible set evaluations; G-QM-Sorkin 0.5-1 and 1-2, driven by triple-path statistics; G-QM-CCR/CAR 1-2 and 3-5, driven by equal-time limit computations; G-QM-MicroCPT/SpinStat 1-3 and 3-6, driven by Minkowski-space tests; G-QM-Tomography/Purification 0.5-2 and 1-3, driven by reconstruction tasks. These are delta additions to the core framework budgets, assuming GPU nodes with at least 16 GB memory.

The kill-switch table for the QM sector is provided here for reference:

QM Claim	Gate(s)	Failure Action
Hilbert space + unitary ($H \geq 0$)	G-QM-OS	Withdraw QM; Euclidean stats only
Born rule	G-QM-Born	No quantum probability; retract QM
Nonlocal but quantum-bounded	G-QM-CHSH/Tsirelson	If ≤ 2 : Classical; if $> 2\sqrt{2}$: Post-quantum; retract
Contextuality (quantum bounds)	G-QM-Context	Retract nonclassical claim
Only 2nd-order interference	G-QM-Sorkin	If order $\neq 2$: Not standard QM; retract
CCR/CAR in continuum	G-QM-CCR/CAR	No canonical quantization; restrict claims
Microcausality + CPT (relativistic)	G-QM-MicroCPT	Limit to nonrelativistic QM on a slice
Spin-statistics (relativistic)	G-QM-SpinStat	Same limitation
Operational reconstruction	G-QM-Tomography/Purification	Downgrade to “some nonclassical statistics,” not

Table 1: Kill-Switch Table for QM Sector

Potential failures in the QM sector are anticipated and handled defensively. For example, if OS reconstruction fails due to insufficient RP or clustering, the immediate action is to stop all QM derivations and retain only Euclidean statistical claims, as clustering is essential for the thermodynamic limit (estimated probability medium 20% in near-critical regions, mitigated by detuning from criticality). If the Born rule anomalies occur, such as violations of contextual additivity, this likely indicates a mismatch in the measurement model (low probability 10%, mitigated by POVM calibration), and the failure is published with retraction of QM claims. Super-quantum CHSH values ($S > 22$) would indicate a post-quantum regime (*medium* 30% in strong-coupling, where entanglement exceeds Tsirelson), and the framework would not be labeled as QM. Non-vanishing third-order interference in Sorkin tests would also signal post-quantum behavior (*low* 15%, mitigated by interference path tuning), leading to retraction. If CCR/CAR do not emerge, possibly due to strong-coupling or finite- a artifacts (medium 35%), the framework seeks a scaling window (e.g., by increasing L or refining smearing functions); if absent, claims are restricted to operational QM without canonical structure. Failures in microcausality or CPT limit claims to nonrelativistic QM on a fixed time-slice (low 20% if Lorentz marginal), provided the nonrelativistic gates (e.g., OS, Born, CHSH) succeed.

5 Gravity Sector: Derivation, Tracks, Gates, and Illustrative Examples

Gravity is derived from the lattice substrate without assuming it a priori, using two complementary tracks to provide robustness and cross-verification.

In Track A, the induced gravity approach (Sakharov-type), anomaly-free matter and gauge fields are integrated out on slow background geometry fields (discrete tetrads e and spin connection), producing a one-loop effective action $\Gamma_{\text{eff}}[e, \Omega] = \int d^4x \sqrt{g} (\Lambda_{\text{ind}} + 1/(16\pi G_{\text{ind}}) R) + \text{higher-derivative terms} + \text{topological terms}$, with positive G_{ind}^{-1} for suitable species content. The lattice heat-kernel expansion uses Seeley-DeWitt coefficients for each species (Dirac DW fermions, scalars, vectors), with the kinetic operator $\mathcal{D}_s(e, \Omega)$ admitting local expansions in covariant curvature invariants due to background slowness and UV gap. Chiral anomalies are controlled by the anomaly gates. RP and gauge covariance are maintained, as backgrounds enter via compact $SO(4)$ holonomies and frame insertions in finite-difference operators. Universal cou-

pling arises from matter interacting only through covariant derivatives and the invariant measure $\sqrt{g}$, with long-wavelength Ward identities reducing to linearized diffeomorphism invariance, enforcing the equivalence principle for the emergent spin-2 field. The soft graviton theorem is tested in the linearized regime, with current conservation and diffeos yielding the standard soft factor at $k \rightarrow 0$. Robustness is ensured by the sub-gate G-GR-EH-ROB, requiring at least three orthogonal curvature families (distinct tensor patterns, each reflection-even), ridge regression with reported λ and Bayesian fits with stated hyperpriors, Leave-One-Background-Out (LOBO) stability index $\geq \sigma_{\text{thr}}$ (preregistered threshold, e.g., 0.8 for variance control), and posterior positivity $Pr(1/(16\pi G_{\text{ind}}) > 0 | \text{data}) \geq 0.99$. For illustration, in a mock regression with fermion-dominant species, $1/(16\pi G_{\text{ind}}) = 1.989$ with SNR 6.2, and higher-derivative ratio 5.5, LOBO stability 0.85. Potential failures include species cancellations yielding wrong sign (mitigated by tuning content, low probability 20% for balanced reps) or non-decoupling higher-derivatives (A4 false, medium 40% near UV, leading to abandonment of Track A).

In Track B, the first-order lattice gravity approach (Palatini/BF route), a compact $SO(4)$ lattice gauge theory for Ω is introduced with a reflection-positive action, coupled to discrete tetrads e via Palatini-type terms or BF plus local quadratic simplicity constraints that are reflection-even and enforce tying e to Ω . RP is verified at the source and post-transformations via G-GR-RP. The linearized sector expands around homogeneous backgrounds ($e = \bar{e} + \delta e, \Omega = \bar{\Omega} + \delta\Omega$), defining a symmetric tensor perturbation $h_{\mu\nu}$ with lattice Ward identities from gauge and simplicity constraints enforcing transverse-traceless conditions in the IR. The propagator structure is fitted to $1/p^2$ with a spin-2 projector at small $|p|$ via G-GR-Spin2. Static conserved sources $T_{\mu\nu}$ (composites from matter) induce a potential $V(r) -G/r$ at large r , with G extracted and universality verified across species. Leakage is bounded by the sub-gate G-GR-Spin2-Leak, quantifying scalar/tensor mixing angle $i \epsilon_{\text{leak}}$ (preregistered, e.g., 0.05) and a null test on scalar-only mock fields requiring observed “spin-2” amplitude $< \epsilon_{\text{null}}$ (e.g., 0.01). In a mock run, spin-2 fit $R^2 = 0.99$, leakage 0.03, null 0.008, EP variance 3.5%, Newton agreement within 1.8σ . Potential failures include no gapless mode (contaminated Goldstones, high 50% in loose constraints, mitigated by retuning couplings) or RP breakdown (low 10%, tighten priors).

The gravity gates are G-GR-RP, which re-runs G-RP with geometric fields included in the transfer operator, requiring all OS inequalities nonnegative within tolerance; failure means no unitary gravitational sector, retaining Euclidean statements only. G-GR-EH performs heat-kernel regression on Γ_{eff} using background curvature modulations, requiring nonzero $1/(16\pi G_{\text{ind}})$ with correct sign and higher-derivative form factors suppressed $\geq 5\times$ in the IR window (defined as momenta $|p| < p_*$ where suppression holds, with p_* quantified from fits, e.g., $p_* = 0.1/a$); with ROB sub-gate as above. Failure means no induced EH, abandoning Track A. G-GR-Spin2 computes the two-point of $h_{\mu\nu}$ or a gauge-invariant proxy at small momenta, projecting onto spin-2 and verifying $1/p^2$ with $R^2 > 0.98$, no spurious poles; with Leak sub-gate. Failure means no propagating graviton, removing gravity claims. G-GR-Ward checks lattice Ward identities linking $p^\mu h_{\mu\nu}$ to sources within CI across small $|p|$; failure breaks diffeomorphism-like symmetry, removing gravity claims. G-GR-EP couples scalar, fermion, vector species as sources, measuring response and requiring species-independent G within errors (variance $< 5\%$); failure means non-universal coupling, removing claims. G-GR-Newton uses static conserved sources to measure $V(r)$ at $r > L/4$ (where finite-size errors are $O(1/L)$ from scaling comparisons), requiring $-G/r$ with slope matching EP within 2σ ; failure means no Newtonian limit, demoting to modified gravity. G-GR-Soft checks IR emission amplitudes for soft factorization at $k \rightarrow 0$ consistent with universality and Wards; failure removes Einstein-like dynamics. G-GR-WW-Formal publishes artifacts documenting WW hypothesis failures, such as $z \neq 1$ or anisotropy i threshold pre-continuum (UV Lorentz absent), lack of Lorentz-covariant gauge-invariant conserved $T_{\mu\nu}$ (constructive counterexample or obstruction proof), or first-order redundancy in Palatini/BF preventing premises; failure means no composite massless spin-2 claim.

Resources for gravity gates: G-GR-EH 2-4 GPU-h at $L=16, 6-10$ at $L=24$, driven by background regressions; G-GR-Spin2 2-3 and 5-8, driven by two-point spectra and projectors; G-GR-Ward/EP 2-4 and 6-10, driven by source insertions; G-GR-Newton 2-4 and 6-10, driven by long-distance potential computations; G-GR-RP folded into core RP; G-GR-Soft 2-4 and 6-10, driven by soft limits.

Theorem G1 (Induced EH Exists), conditional on G-GR-EH: If the gate passes, the EH term with correct sign is present in Γ_{eff} , and higher-derivative terms are subleading in the IR window (quantified as suppression $\geq 5\times$ for $|p| < 0.1/a$, with errors $O(a^2)$ from lattice artifacts). Proof: Constructive via regression on orthogonal backgrounds; measured positivity from Bayesian posteriors; suppression from fitted

magnitudes and scaling analyses comparing different a .

Proposition G2 (Massless Spin-2 Propagation), conditional on G-GR-Spin2: If the spin-2 projector component scales as $1/p^2$ at small p , the IR contains a massless helicity-2 excitation. Proof: Standard spectral analysis on the lattice identifies a zero-mass pole in the spin-2 channel, with the projector eliminating scalar or vector contamination; leakage bounds ensure purity.

Proposition G3 (Universal Coupling = Equivalence Principle), conditional on G-GR-Ward and G-GR-EP: If linearized diffeomorphism Ward identities hold and all tested matter species couple to $h_{\mu\nu}$ with the same IR G (variance $< 5\%$), the equivalence principle holds at leading order.

Theorem G4 (Newtonian Limit), conditional on G-GR-Newton: If static sources generate $V(r) = -G/r$ with G consistent across species and the spin-2 pole exists, the IR realizes the Newtonian potential as the static limit of massless spin-2 exchange, with finite-size errors $O(1/r^2)$ for $r > L/4$.

6 Chemistry Sector: CTMC-Rigorous and Parametric Claims with Expanded Sensitivities

The chemistry sector models an open chemical system using the chemical master equation (CME) with inflows λ_i , dilution δ , and favorable rates during fraction ϕ of cycles. All claims are parametric to allow for environmental variations, with sensitivities analyzed for $\pm 20\%$ perturbations.

Claim C1 (Reachability with Dilution): For a reaction path of length r from inflows to an activated ribonucleotide, with per-step rate $\geq k_{min}$ during ϕ , the probability of success by time τ is $\geq 1 - \exp(-\Lambda\tau)$, where $\Lambda \geq c\phi(k_{min})^r \exp(-\delta T)$, with cycle duration T and constant $c > 0$ path-dependent. For base parameters $\phi = 0.2$, $k_{min} = 10^{-3}s^{-1}$, $\delta = 0.05s^{-1}$, $T = 10s$, and $r = 6$, $\Lambda \approx 10^{-19}s^{-1}$; sensitivity to k_{min} is dominant ($\Lambda \propto (k_{min})^r$, so $\pm 20\%$ changes Λ by factors of $(1.2)^6 \approx 3$ and $(0.8)^6 \approx 0.26$); to $\phi \pm 20\%$ linear scaling; to δ exponential in $-\delta T$, $\pm 20\%$ changes $\exp$ term by $\exp(\pm 1)$.

Claim C2 (Polymerization Persistence): For ligation ℓ , supply s , cleavage c , dilution δ , if $\ell s > c\delta$, then for Lyapunov function $V(L) = L$ or L^2 , there exist ϵ , L_* , C such that $\mathcal{L}V(L) \leq -\epsilon 1_{L \geq L_*} + C$, implying positive recurrence and nonzero stationary mass on long polymers (Meyn-Tweedie theorem). For base $\ell = 0.10$, $s = 2.0$, $c = 0.05$, $\delta = 0.05$, the inequality holds with margin 0.10, yielding certificate ($\epsilon = 0.01$, $L_* = 20$, $C = 4.3$) for $V = L^2$; sensitivity to $\ell \pm 20\%$ scales ϵ linearly and shifts L_* mildly (to 16/24); to s similar; to c or δ opposite, with failure if margin < 0 .

Claim C3 (Eigen Threshold): For per-site fidelity $q = 1 - \mu$, length L , replication rates $r_* > r$, compartmentalization $1 + \eta$, the master quasispecies has nonzero stationary weight if $q^L > (r/r_*)/(1 + \eta)$. For base $q = 0.995$, $r_*/r = 1.5$, $\eta = 0.2$, RHS ≈ 0.556 , allowing L up to 116; sensitivity to $q \pm 20\%$ on error rate $(1 - q)$ reduces/increases max L linearly in $\log$ scale (to 93/139); to η increases relax threshold, allowing $+20\%$ larger L ; to r_*/r opposite.

Expanded cases: For $q = 0.99(\pm 20\% \text{ error to } 0.008/0.012)$, max $L = 69(56/82)$; for $k_{min} = 10^{-4}$ in C1, Λ drops to 10^{-25} , highly sensitive. Additional: for $\phi = 0.1$, Λ halves; for $r = 8$, Λ drops by $(k_{min})^2$.

The G-CHEM gate publishes (ϵ, L_*, C) , Λ , Eigen check, and sensitivities. Failure withdraws RNA claim for the environment.

7 Simplicity-Based Selection: Structures, Parameters, and Gate

The candidate class is a countable set of anomaly-free, BEH-feasible chiral gauge theories with one unbroken $U(1)$, enumerated with $\text{rank}(G) \leq 4$, irrep $\dim \leq 6$, ≤ 12 Weyl flavors per factor, justified by computational cost while covering SM-like models (rationale: these bounds limit enumeration to 100 candidates, feasible for MDL computation, while rank 4 includes $SO(10)$ GUTs and dim 6 covers fundamentals/adjoint).

Discrete structures have prefix lengths l_α satisfying the Kraft inequality for prefix codes, with priors $\pi_\alpha \propto 2^{-l_\alpha}$; parameter priors are proper, allowing two-part codes for flexibility. The evidence is $\mathcal{Z}_\alpha = \int L_\alpha(\theta) \pi_\alpha(\theta) d\theta$, and MDL is $-\log L_\alpha(\hat{\theta}_\alpha) + (d_\alpha/2) \log n + C_\alpha$, where d_α is parameter dimension, n sample size, C_α complexity term.

Scoped consistency S1: If the true model is in the class, identifiable with positive Fisher information at $\hat{\theta}$ under standard regularity, MDL is consistent for model selection.

The identifiability sheet per candidate includes: structure G_α , representations $\mathcal{R}_\alpha$, charges; observables used (e.g., momenta, volumes, BEH diagnostics); parameterization θ with priors and bounds; regularity conditions; empirical Fisher matrix $\hat{I}(\hat{\theta})$ with min eigenvalue $\geq \lambda_{min}^{thr}$ (preregistered, e.g., 0.1 for invertibility); degeneracy handling via regularization (e.g., ridge if ill-conditioned); MDL/evidence numbers; and rationale for inclusion/exclusion.

The G-SELECT gate publishes the candidate list, prefix lengths, priors, $(\alpha1, \alpha2, \alpha3)$ grid for simplicity functional $\mathcal{C} = \alpha1K + \alpha2 \text{ params} + \alpha3 \text{ forbidden Yukawas}$, and top-k rankings. Failure means explicit design choice for SM, without “emergent” language.

8 Unification: Overlap Optimization, Readiness Playbook, and Expanded Risks

Unification requires a shared parameter window where the QM gate bundle (G-QM-OS, G-QM-Born, G-QM-CHSH/Tsirelson with causality, G-QM-Sorkin, G-QM-CCR/CAR, optional MicroCPT/SpinStat) and gravity bundle (G-GR-EH with ROB or G-GR-Spin2 with Leak + G-GR-EP + G-GR-Newton; G-GR-RP; G-GR-WW-Formal) pass concurrently. The IR window for unification is quantified as distances $r > L/4$, where finite-size errors scale as $O(1/L)$ based on comparisons between $L = 16$ and $L = 24$ runs, with error estimates from bootstrap resampling of correlators (e.g., $O(5\%)$ at $L = 24$).

The G-OVERLAP-OPT gate performs scripted grid search or Bayesian optimization over parameters $(m, \varepsilon_{adm}, \tau, L_5, \text{block size})$, with hard constraints that all core gates pass, and a soft score as the sum of pass indicators for the QM bundle and one gravity track minus penalties for CI widths and leakage values. Output includes the best parameter slice plus top-k alternates, with full gate cards. For illustration, in a mock optimization, the best slice was $\{m : 0.52, \varepsilon_{adm} : 0.1, \tau : 0.06, L_5 : 12, \text{block: '23'}\}$, with joint score 8.5 out of 10, CHSH 2.6 ± 0.01 , and spin-2 $R^2 0.99$. Failure reports “disjoint windows” as a publishable negative.

The run-order is fail-fast: core gates first, then QM quicks (OS, CHSH with causality, Sorkin) to kill post-quantum or classical early, gravity fork in parallel (A with ROB, B with Leak), QM completion (Born, CCR/CAR), WW-Formal, overlap optimizer, optional Lorentz then MicroCPT/SpinStat.

Key performance indicators (KPIs) include: RP min margin $\geq -10^{-8}$ with CI $< 3 \times 10^{-9}$; CHSH S in $(2, 2\sqrt{2})$ with CI < 0.03 and no-signaling margins $> LR$ bound; Sorkin $I_3 \approx 0$ within bootstrap CI (e.g., ± 0.01); Born residuals $\leq 1.5\sigma$; CCR error $\propto \alpha^a$, $\alpha > 0.8$; EH posterior positivity ≥ 0.99 , LOBO ≥ 0.8 , HiDer $\geq 5\times$; Spin2 leakage < 0.05 , null < 0.01 , $R^2 > 0.98$; EP variance $< 5\%$; Newton linear for $r > L/4$, match 2σ .

The expanded risk register includes probability estimates (low $< 10\%$, medium $10 - 50\%$, high $> 50\%$) based on prior simulations:

Risk	Symptom	Mitigation	Probability Estimate	Chained Failure Scenario
RP erosion post-RHK	OS margin dips negative at small τ	Reduce τ , symmetrize stencils, re-seed; if persists, tighten ε_{adm}	Medium	If fails after retune (20% chance), chains to no unitarity, affecting 80% of QM/GR claims
Anomaly leakage	Non-zero phase on some π_1 loop	Reduce Q_{max} ; adjust reps; re-cool configs; if persistent, redesign matter	Low	Persistent (10% chance) chains to restricted $\mathcal{R}$, reducing unification window by 50%
Post-quantum stats	CHSH $> 2\sqrt{2}$ or Sorkin $I_3 \neq 0$	Declare “post-quantum”; withdraw QM claim for slice	Medium	If supra-quantum (30% in strong-coupling), chains to full QM retraction, blocking unification
No spin-2 pole	Projected 2-pt behaves massive	Retune Palatini/BF couplings; try different blocks; abandon B	High	Fail after retune (60% chance) chains to rely on A only, reducing gravity scope by 50%
No induced EH	Regression unstable or wrong sign	Increase species; alter UV cutoffs; abandon A	Medium	Wrong sign (40% if species poor) chains to B only, with 20% risk of joint failure
Non-universal coupling	G differs by species	Check Wards; if persists, no Einstein claim	Low	Persistent (15% chance) chains to modified gravity demotion, affecting unification claim
CCR/CAR stall	No convergence across $L = 16 \rightarrow 24$	Increase L or smearing; restrict to operational QM	Medium	No window (35% in finite- a) chains to non-canonical QM, weakening bundle by 30%

Table 2: Expanded Risk Register

Decision tree and reporting templates as before, with added fields for KPIs (e.g., “CHSH CI: 0.02”).
Resources delta to core: QM quicks 2-3 GPU-h $L = 16$, 5-6 $L = 24$; gravity A 2-4, 6-10; B 6-11, 12-20; CCR 1-2, 3-5.
Success criteria: Provisional IR unification if bundles pass shared window (e.g., $r > L/4$, error $O(1/L)$), WW OK; optional relativistic alignment. Claim wording as before.
Demotion paths: QM fail $\rightarrow$ Euclidean; gravity partial $\rightarrow$ induced or spin-2 only; WW $\rightarrow$ no composite; Lorentz fail $\rightarrow$ nonrelativistic.
Quick-start YAML updated with `sorkintolspecified`.

9 Claims to Gates: Complete Kill-Switch Table

Claim	Required Gates	Failure Action
Unitary QFT via OS	G-RP all stages, clustering	Withdraw unitarity; Euclidean stats
Chiral fermions no doublers	G-GW	Drop chirality or retune
Anomaly freedom incl mixed	G-ANOM, G-MIXED-ANOM	Drop claim/scope
Admissibility preserved	G-ADM + re-G-RP	Abort/retune/declare scope
Emergent Lorentz	G-LORENTZ	Remove Lorentz claims
Born rule	G-QM-Born	No quantum probability; retract QM
Quantum nonlocality bounded	G-QM-CHSH/Tsirelson, G-QM-CHSH-Causality	Classical or post-quantum
Sorkin order-2 only	G-QM-Sorkin	Post-quantum or classical; retract QM
CCR/CAR continuum	G-QM-CCR/CAR	Restrict to operational QM
Induced EH	G-GR-EH, G-GR-EH-ROB	No EH claim
Massless spin-2	G-GR-Spin2, G-GR-Spin2-Leak	No spin-2 claim
Equivalence Principle	G-GR-EP	No universal coupling
Newtonian limit	G-GR-Newton	No Newtonian
WW compliance	G-GR-WW-Formal	No composite spin-2
IR Unification QM $\wedge$ GR	G-OVERLAP-OPT + QM bundle + GR-A or GR-B	Publish disjoint-window
RNA feasibility	G-CHEM	Withdraw for environment
Emergent SM	G-SELECT	Design choice: SM

Table 3: Complete Kill-Switch Table

10 Runbook, Artifacts, Resources, Minimal Instantiations

Artifact layout as before, with added unification/ for mock slices (e.g., JSON with params and KPIs).

Resources: Core G-RP 2-4 GPU-h $L=16$, 6-10 $L=24$; GW 1-2, 3-5; ADM 0.5-1, 2-3; ANOM 3-6, 8-14; DECIDE 10-30 min CPU; LORENTZ 2-4, 6-10; CHEM seconds.

Minimal instantiations: $SU(2)L = 16/24$, DW $L5 = 12$, $m = 0.48/0.52/0.56$, $\varepsilon_{adm} = 0.10$, blocks 2^3 , $\tau = 0.04/0.06/0.08$, $Q_{max} = 1$ (2 test). Targets: $\xi \leq 2.0$ (CI), residual $< 8e - 3$, phases $< 1e - 10$, treewidth ≤ 12 .

11 Scope, Open Items, Defensive Engineering

Scope: Instance-verified, limited to tested windows (e.g., $L \leq 24$, $Q \leq 1$), with IR as $r > L/4$, errors $O(1/L)$. Open: H-YM mass gap beyond strong-coupling (tagged, low probability $< 10\%$ in operational windows); Conjecture-Sel SM simplicity (tested, medium 30% chance of non-preference). Relativistic requires G-LORENTZ; otherwise nonrelativistic.

Defensive: Transformations proved/re-tested; failures localized (e.g., chained probabilities in risks). Outcomes decisive for publication.

12 Technical Appendix: Pseudocode and Implementations

This appendix consolidates all pseudocode for gates, avoiding repetition in main text.

12.1 G-RP Pseudocode

```
for stage in ["pre", "post_block", "post_RHK"]:
    margins = []
    \sigma = pilot_std(os_cone_basis, stage, n0=200, antithetic=True)
    n = ceil(2*\sigma**2 / \varepsilon**2 * log(1/\alpha)) * 1.2
    for F in os_cone_basis:
        vals = [inner(F, T(stage) @ \Theta(F), sample=i) for i in 1..n]
        \mu, ci = mean_CI(vals)
        margins.append((\mu, ci))
    assert min(m.\mu for m in margins) >= -\varepsilon
    save_json(stage, margins, n, \sigma, \alpha, correction="Bonferroni_or_BH")
```

12.2 G-GW Pseudocode

```
xi_CIs = []
for cfg in canonical_edge_generators(group, eps_adm):
    dists, norms = sample_kernel_norms(DW_operator, cfg, probes=K)
    xi_hat, ci = fit_exponential_tail(dists, norms, robust=True) # e.g., Theil-Sen
    xi_CIs.append(ci)
assert max(ci.upper for ci in xi_CIs) <= L/8
gw_res = estimate_GW_residual(DW_operator, cfg_pool, probes=200) # Hutch++
assert gw_res <= gw_residual_threshold
```

12.3 G-ADM Pseudocode

```
for \tau in tau_grid:
    U_block = apply_block_map_B(U)
    U_post = apply_RHK(U_block, \tau, symmetric_stencil=True)
    max_dev = max_plaq_deviation(U_post)
    assert max_dev <= eps_adm
    run_G_RP(stage=f"post_RHK_tau{\tau}")
```

12.4 G-ANOM / G-MIXED-ANOM Pseudocode

```
for Q in range(-Qmax(L), Qmax(L)+1):
    for cfg in base_configs(Q, cooled=True, admissible=True):
        for loop in loop_basis(cfg): # contractible + \pi generators
            \phi_g = fujikawa_phase(cfg, loop, mixed=False)
            \phi_m = fujikawa_phase(cfg, loop, mixed=True, spin_conn=\omega)
            assert is_zero(\phi_g) and is_zero(\phi_m) # to tolerance
            log_phase(cfg_id, loop_id, \phi_g, \phi_m)
```

12.5 G-DECIDE Pseudocode

```
graph = interaction_graph(post_block_code)
tw, cert = tree_decompose(graph)
assert tw <= w_max
assert no_adaptive_control(ir)          # static analysis
hits = scan_forbidden_templates(graph, patterns)
assert len(hits) == 0
save(cert, hits)
```

12.6 G-LORENTZ Pseudocode

```
z1, ci1, c1 = fit_dispersion_small_p(spectral_data)
z2, ci2, c2, \chi2 = finite_size_collapse(data)
assert overlaps(ci1, [1-\delta, 1+\delta]) and overlaps(ci2, [1-\delta, 1+\delta])
assert abs(z1 - z2) <= \delta/2
y_aniso, ci = fit_anisotropy_exponent(data)
assert ci.upper < 0
```

12.7 G-QM-Born Pseudocode

```
# Given POVM {E_j} and sub-POVM refinements producing same projector sums:
for frame in frames:
    p_sum = sum( freq(E) for E in frame.elements )
    # Compare across different refinements that share the same projector
    assert abs(p_sum - target_prob(frame.projector)) <= tol
# Fit density operator rho via tomography; check p(E)=Tr[rho E] within CI.
```

12.8 G-QM-CHSH/Tsirelson Pseudocode

```
S = estimate_CHSH(correlators) # S = E(a,b)+E(a,b')+E(a',b)-E(a',b')
assert S > 2 - eps # Bell violation
assert S <= 2*sqrt(2) + eps # Tsirelson cap
```

12.9 G-QM-Sorkin Pseudocode

```
I_ABC = intensity(ABC)
I_AB = intensity(AB); I_BC = intensity(BC); I_CA = intensity(CA)
I_A = intensity(A); I_B = intensity(B); I_C = intensity(C); I_0 = intensity(empty)
I3 = I_ABC - I_AB - I_BC - I_CA + I_A + I_B + I_C - I_0
assert abs(I3) <= tol # tol from bootstrap CI
```

12.10 G-QM-CCR/CAR Pseudocode

```
# Choose test functions f,g; approximate equal-time commutators via time-slice
  limits
C = commutator( phi(f,t), pi(g,t) )
assert close_to_i_delta(C, tol_scaling_with_a)
F = anticommutator( psi(f,t), psidag(g,t) )
assert close_to_delta(F, tol_scaling_with_a)
```

12.11 G-QM-MicroCPT Pseudocode

```
# Microcausality: spacelike separated supports
comm = commutator( O(x), O(y) ) # extract from reconstructed Minkowski 2-pt
assert within_CI_zero(comm)
# CPT: compare correlators under CPT-transformed configurations
assert correlators_equal_within_CI(CPT_apply(data))
```

12.12 G-CHEM Pseudocode

```
(\varepsilon, Lstar, C) = lyapunov_certificate(\ell, s, c, \delta, V=L^2 or L)
\Lambda = reachability_rate(k_min, \phi, \delta, T, r)
assert \ell*s > c*\delta
assert q*L > (r/r_star) / (1+\eta)
publish({"epsilon": \varepsilon, "Lstar": Lstar, "C": C, "Lambda": \Lambda})
```